Federico Pardo García

AI Engineer | PhD in Computer Science & AI, specialized in Explainable AI
EU Citizen | Relocation Ready | Immediate Availability
(+34) 600 969 768 | fpardo.net | federico.pardog@gmail.com | LinkedIn | GitHub

Profile

PhD in Computer Science & AI (Summa Cum Laude) specialized in **explainable AI**, with three first-author publications in IEEE Access and Applied Sciences on interpretability, multimodal AI and domain adaptation. My research reaches production rather than stopping at a paper: I operationalized **SHAP-based interpretability over fine-tuned BERT models** inside MoviSound, a live platform serving real users, so that model outputs were transparent and auditable rather than black-box scores. Five years of engineering experience turning frontier methods into systems that run, including 8 months supporting enterprise clients on Google Cloud's data and AI stack (BigQuery, Vertex AI).

Education

University of Murcia — Murcia, Spain
PhD in Computer Science and Artificial Intelligence ***(Summa Cum Laude)*** — *May 2023 – May 2026*
- Thesis: *Enriched Feedback of Classroom Dynamics Using AI.* Specialization in explainability and interpretability of multimodal models, and in domain adaptation for generalizable model behaviour across heterogeneous populations.

University of Murcia — Murcia, Spain
M.Sc. in Big Data and Artificial Intelligence (GPA: 8.7/10) — *2020 – 2021*

University of Murcia — Murcia, Spain
B.Sc. in Computer Science, Specialization in AI — *2016 – 2020*

Publications

Five peer-reviewed publications, three as first author, on interpretability, multimodal AI and domain adaptation. Active in the research community and open to shaping a publication agenda.

Pardo, F. (2025). *Explaining Teacher Interventions in SRS-Based Classrooms.* IEEE Access. Interpretability of model decisions in deployed settings.

Pardo, F. (2025). *Audio Features in Education: A Systematic Review.* Applied Sciences, 15(12), 6911.

Pardo, F. (2024). *Exploring AI Techniques for Generalizable Teaching Practice Identification.* IEEE Access. Model generalization and domain adaptation.

Pardo, F. et al. (2024). *Analyzing Wooclap's Competition Mode with AI Through Classroom Recordings.* IEEE RITA. (Co-author)

Pardo, F. et al. (2023). *AI-Driven Teacher Analytics: Informative Insights on Classroom Activities.* IEEE TALE. (Co-author)

Experience

AI Engineer — May 2023 – May 2026
University of Murcia — *Murcia, Spain*
- **Operationalized interpretability in production**: implemented SHAP-based Explainable AI frameworks over fine-tuned BERT models, turning opaque predictions into transparent, verifiable feedback that end users could act on and challenge.
- Shipped MoviSound, a live authenticated platform serving real users, built on end-to-end multimodal pipelines integrating ASR (Whisper), speaker diarization, paralinguistic feature extraction, and semantic embeddings with vector search (FAISS).
- Designed and operated a decoupled microservices architecture (Python, FastAPI, Docker, Celery) for multimodal inference, ensuring reliability and high availability across local and cloud environments.
- **Domain adaptation and model evaluation**: researched and benchmarked techniques for generalizable model behaviour across heterogeneous populations, published as a first-author IEEE Access paper.
- Published a systematic review and 4 empirical papers, translating research methodology into deployed engineering solutions.

Cloud Technical Support Specialist (GCP) — Aug. 2022 – Apr. 2023
Webhelp (Google Cloud Project) — *Barcelona, Spain*
- Provided L2 engineering support for enterprise clients on Google Cloud's **data and AI stack**, covering BigQuery and Vertex AI alongside core infrastructure (Compute Engine, VPC Networking, IAM, Google Kubernetes Engine).
- Debugged production issues in distributed Linux environments with SRE teams, working entirely in English with international enterprise customers under incident-response SLAs.

Data Scientist and Machine Learning Engineer June 2021 – Aug. 2022
CENTIC (Technological Center) *Murcia, Spain*

- Deployed a real-time CNN for industrial defect detection at 98% accuracy with sub-100ms inference latency on embedded hardware, validated against production constraints before rollout.
- Implemented and optimized an LSTM-based NLP architecture for linguistic complexity evaluation in resource-constrained inference environments.

Projects

Sextante | *Next.js, NestJS, TypeScript, PostgreSQL, Docker, MCP, OAuth 2.1* | sextante.fpardo.net 2026 – Present

- Built, deployed and operate a production personal-finance platform for the Spanish market: portfolio aggregation with live quotes and IRPF, wealth, gift and Social Security calculators against real 2026 tax brackets, where correctness is regulatory rather than approximate.
- Ships an MCP (Model Context Protocol) server secured with OAuth 2.1, exposing the platform as a governed, authenticated tool integration for LLM assistants.

MoviSound | *Python, FastAPI, Docker, Celery, Whisper, BERT, SHAP, FAISS* | movisound.inf.um.es 2023 – 2026

- Multimodal AI platform giving teachers objective, automated and explainable feedback from classroom recordings. Deployed as an authenticated web platform with real users; the engineering output of the doctoral research.

Technical Skills

Responsible & Explainable AI: SHAP (XAI), interpretability of fine-tuned transformers, model evaluation and benchmarking, domain adaptation and generalization, transparent production deployment of research models.
Generative AI & Frameworks: LLMs (Llama, Mistral), Hugging Face Transformers, BERT fine-tuning, Whisper (ASR), Speaker Diarization, Vector Search (FAISS), Multimodal Fusion, Vertex AI, MCP (Model Context Protocol).
ML & Data Engineering: PyTorch, TensorFlow, ETL pipelines, BigQuery, Signal Processing (Librosa), Feature Engineering at scale, CNNs, RNNs (LSTM).
Systems & Infrastructure: Docker, Kubernetes (GKE), FastAPI, Celery, Linux, Unraid, GCP (BigQuery, Vertex AI, Compute Engine, VPC, IAM), OAuth 2.1, Ollama (local LLM serving).
Programming Languages: Python (Expert), C++, TypeScript, SQL, Bash, JavaScript, Go (working knowledge).
Languages: Spanish (Native), English (C1 Advanced), German (Elementary).